

The Planet Earth

GS 21/L: Physical and Structural Geology

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Chapter 2 Overview



2.1 Origins

Theories explaining the formation of the universe, our solar system, and the chemical differentiation that built early Earth.



2.2 Size & Shape

Understanding Earth's true dimensions, the oblate spheroid concept, and the forces causing its equatorial bulge.



2.3 & 2.4 Structure

Analyzing Earth's internal compositional and physical layers, and exploring the concept of isostasy and crustal equilibrium.

Earth's Major Spheres

The Four Spheres

Atmosphere: The gaseous envelope supporting life and protecting the surface from radiation.

Hydrosphere: The dynamic mass of water continually on the move. Oceans cover 71% of Earth's surface; 97% is saline water.

Geosphere: The thickest of the spheres, extending nearly 6,400 km downward to the core.

Biosphere: Comprises all life found on Earth, deeply integrated with the other three spheres.



2.1 Origin of the Earth & Solar System

The Big Bang Theory

- Began approximately **13.7 to 13.8 billion years ago** with a massive expansion that sent matter flying in all directions.
- As the initial superheated debris expanded and cooled, subatomic particles coalesced into the first elements (primarily hydrogen and helium).
- Gravitational collapse of massive gas clouds initiated fusion reactions, forming the first primitive stars and galaxies.
- *Cosmic Microwave Background (CMB)* radiation serves as the primary modern academic evidence for this rapid initial expansion.



The Nebular Theory



Solar Nebula

Our solar system originated from a massive, rotating cloud of interstellar gas and dust (primarily hydrogen and helium) known as a solar nebula.



Gravitational Collapse

Shockwaves, possibly from a nearby supernova, caused the nebula to contract. It began spinning faster, flattening into a protoplanetary disk.

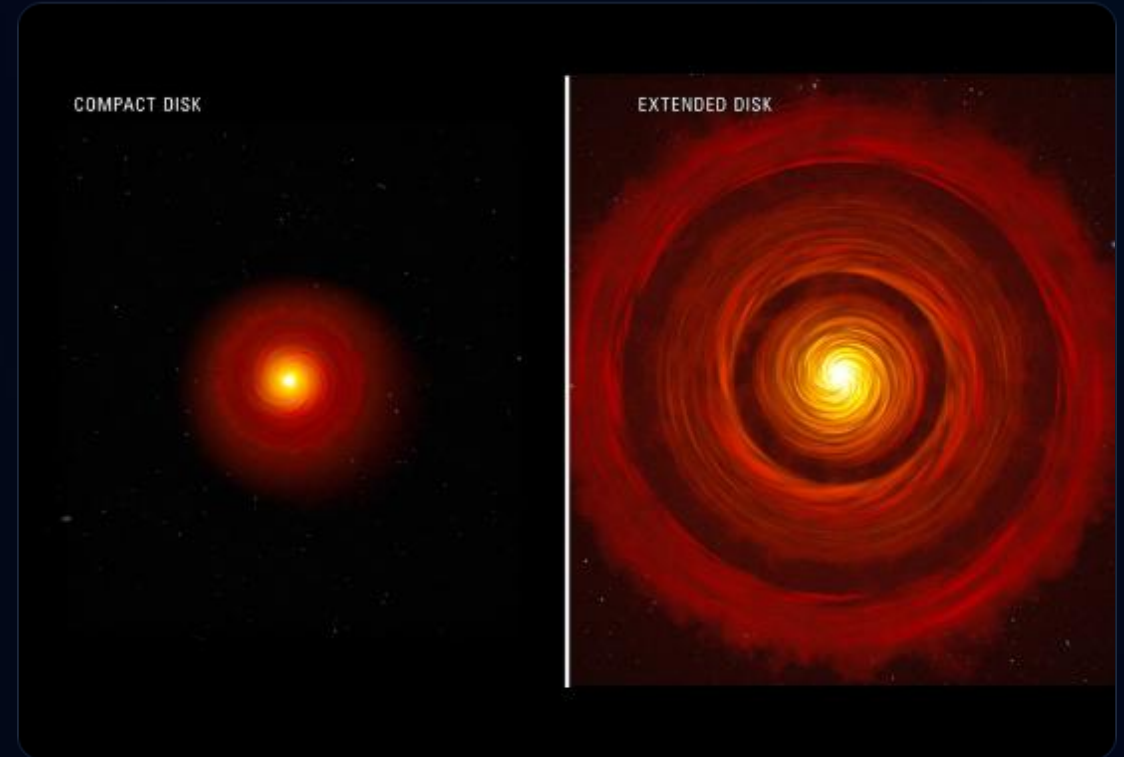


Protostar Ignition

The vast majority of mass concentrated at the center. Immense pressure and gravitational heating triggered nuclear fusion, igniting the young Sun.

Formation of the Planets

- **Inner (Terrestrial) Planets:** Near the Sun, temperatures were too high for ices/gases to condense. Particles with high melting points (metals, silicates) coalesced into rocky planets.
- **Outer (Jovian) Planets:** Beyond the "frost line," cooler temperatures allowed massive quantities of water, ammonia, and methane ices to accumulate.
- **Accretion:** The process where colliding particles fuse together over millions of years, growing from dust to planetesimals, and finally to full-sized planets.



Chemical Differentiation



The Iron Catastrophe

During the Hadean Eon, early Earth experienced intense heating from radioactive decay and high-velocity impacts, creating a global magma ocean.

Density drove differentiation: heavy metals like iron and nickel sank toward the center to form the Core. Lighter, buoyant elements (silicon, aluminum) rose, eventually cooling to form the primitive Crust and primitive atmosphere.

2.2 Size and Shape of the Earth

The Oblate Spheroid

21km

Equatorial Bulge Difference

Centrifugal Force at Work

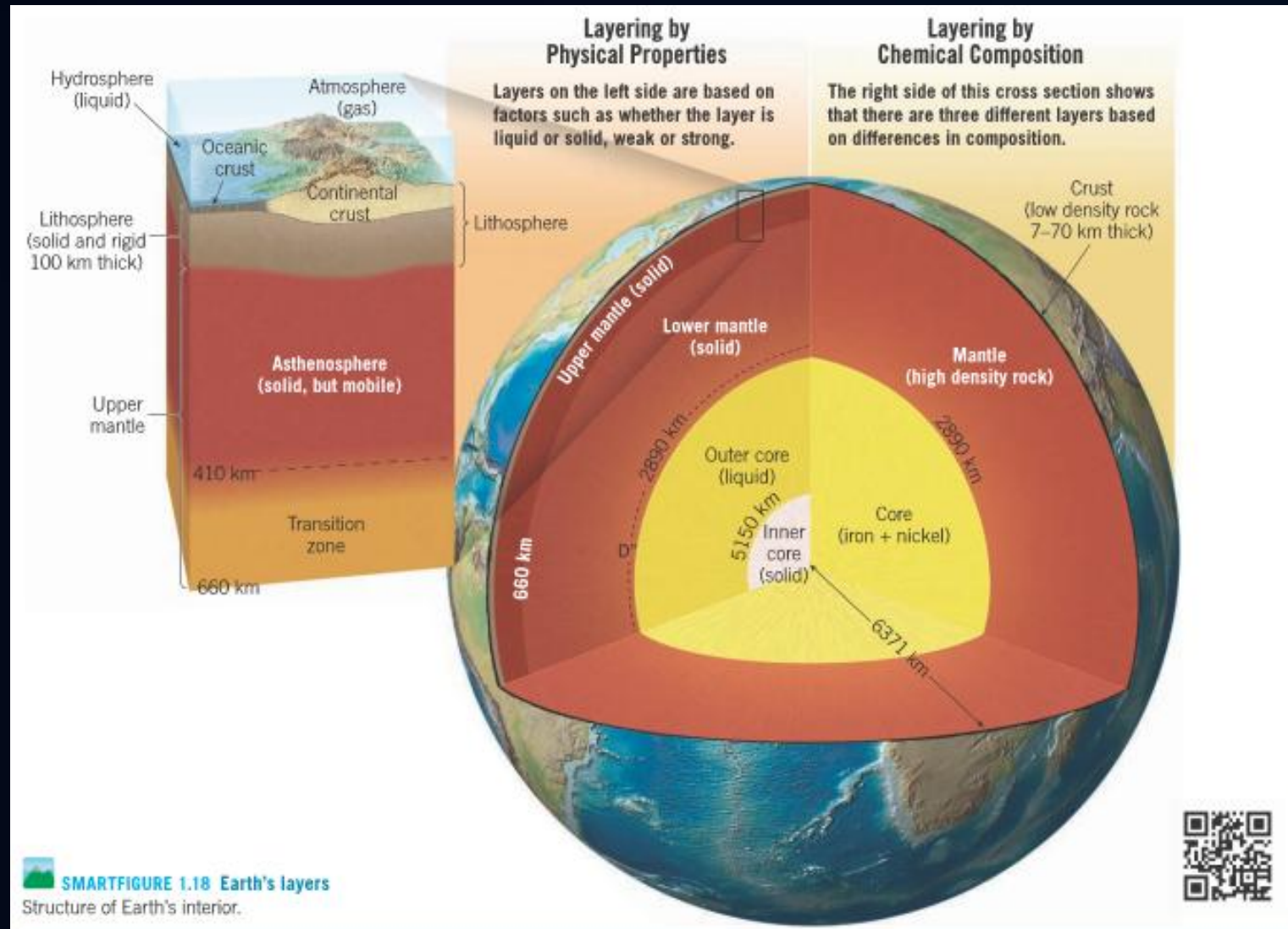
Earth is not a perfect geometric sphere. Its rotation generates centrifugal force, causing the planet's mass to bulge at the equator and flatten at the poles.

The equatorial radius is approximately 6,378 km, while the polar radius is about 6,357 km. In modern geodesy, this shape is mathematically defined as a reference ellipsoid (e.g., WGS 84).

2.3 Earth's Internal Composition

Compositional Layering

- **Crust:** The low-density outermost skin. Divided into thick Continental Crust (granitic, density $\sim 2.7 \text{ g/cm}^3$) and thin Oceanic Crust (basaltic, density $\sim 3.0 \text{ g/cm}^3$).
- **Mantle:** A solid, rocky shell extending to 2,900 km, comprising 82% of Earth's volume. Rich in iron and magnesium silicate minerals (peridotite).
- **Core:** The dense central sphere composed predominantly of an iron-nickel alloy, with minor amounts of oxygen, silicon, and sulfur.



Physical Property Layering

Outer Layers

Lithosphere: The rigid outer shell (~100km thick), containing the crust and uppermost mantle. It is broken into tectonic plates.

Asthenosphere: A weak, ductile (plastic) layer of the upper mantle beneath the lithosphere, allowing tectonic plates to slide.

Mesosphere: The strong, solid lower mantle extending down to the D" layer at the core boundary.

The Core Regions

Outer Core: A liquid iron-nickel layer. The convective flow of this electrically conductive fluid generates Earth's magnetic field (Geodynamo effect).

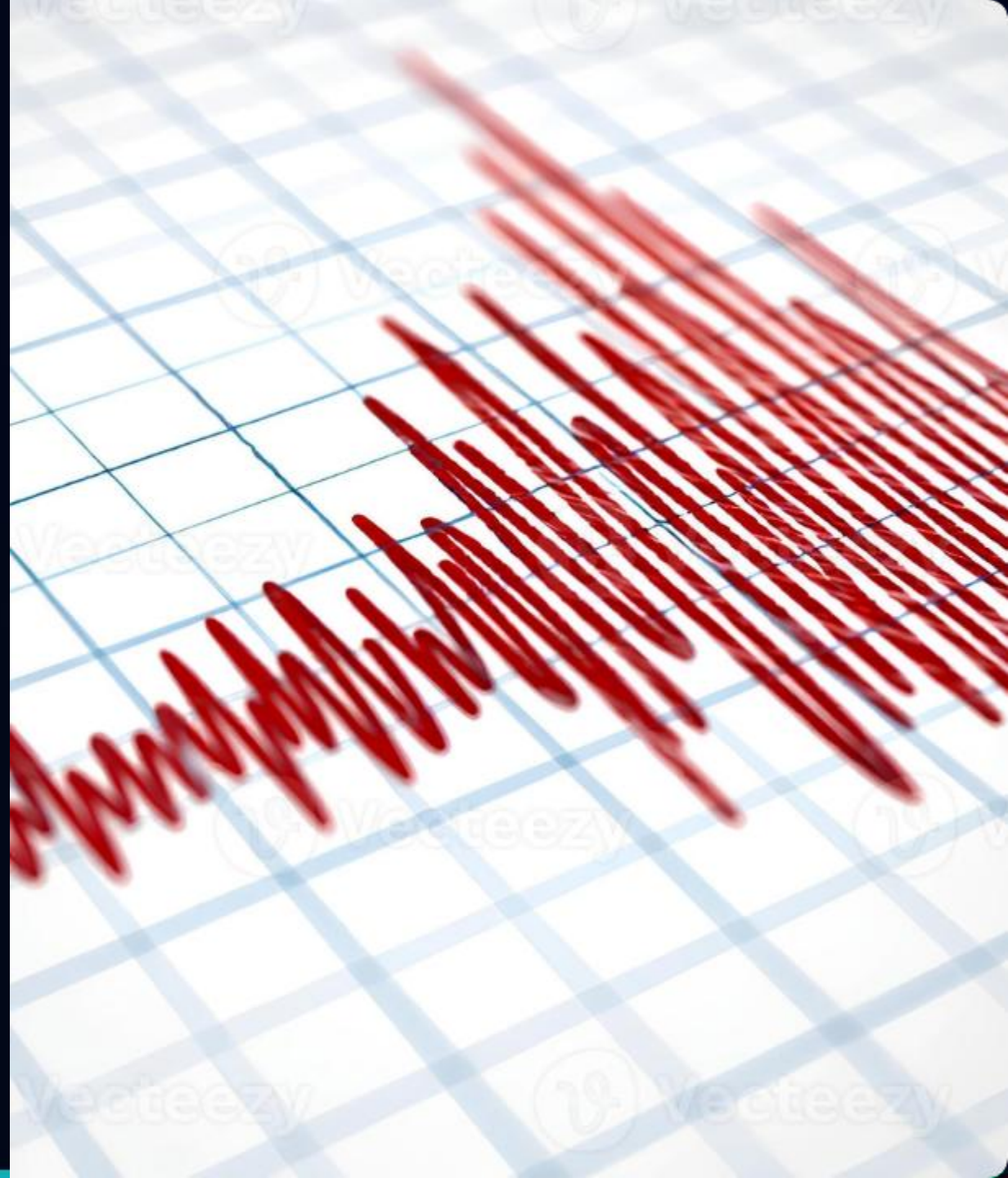
Inner Core: Despite extreme temperatures (comparable to the Sun's surface), immense confining pressure keeps the iron-nickel alloy in a solid state.

Probing the Interior

Because we cannot drill into the mantle or core, our knowledge of Earth's interior comes primarily from **Seismology**.

Seismic waves (P-waves and S-waves) change velocity, refract, or reflect when they encounter materials of different densities and states.

For example, S-waves cannot travel through liquids, creating a "shadow zone" that definitively proved the Outer Core is liquid.



2.4 The Concept of Isostasy

Defining Isostasy

“ The tendency of the Earth's crust to seek equilibrium... a condition of ideal balance. ”

– Clarence Edward Dutton (American Seismologist & Geologist)

Early Isostatic Hypotheses

Airy's Model (1855)

Proposed that the crust possesses uniform density but *varying thickness*. Similar to icebergs, thicker crustal blocks (mountains) have deeper "crustal roots" projecting down into the denser mantle to support their elevated topography.

Pratt's Model (1859)

Proposed that the crust possesses *varying densities* but reaches a uniform base depth (Depth of Compensation). Mountains extend higher because they are composed of lower-density rock compared to the surrounding crust.

Modern Isostatic Concepts

- **Heiskanen's Model:** Proposed a pragmatic compromise blending both varying crustal thickness (Airy) and varying crustal density (Pratt).
- **Hayford-Bowie Concept:** Currently the most accepted model in geodesy. Proved that gravity anomalies relate directly to topographic features.
- They established the formal mathematical concept of the **depth of isostatic compensation**, essential for understanding tectonic uplift and subsidence.



Glacial Isostatic Adjustment

- Isostasy is not a static concept; it responds to changing surface loads.
- **Isostatic Rebound:** When a massive load is removed (like the melting of continental ice sheets after the last Ice Age), the crust slowly rises back to equilibrium.
- Regions like Scandinavia and Hudson Bay are still experiencing measurable upward crustal rebound today as the viscous mantle flows back underneath them.



QUESTIONS?

References



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